

Vital Sign Description: Respiratory Rate

Respiratory rate is a fundamental vital sign that measures the number of breaths a person takes per minute. It is one of the five primary vital signs, alongside pulse rate, blood pressure, temperature, and oxygen saturation (Cretikos et al., 2008). Respiratory rate is assessed by observing the rise and fall of the chest wall over a 60-second period, counting each complete inhalation and exhalation cycle. In clinical practice, it provides critical information about a patient's respiratory function and overall physiological status, making it an essential component of emergency triage assessment (Flenady et al., 2017).

The clinical range for respiratory rate varies by age group, with normal adult values typically between 12 and 20 breaths per minute (bpm) (World Health Organization, 2019). For adults, a respiratory rate below 12 bpm is classified as bradypnea (abnormally slow breathing), while a rate above 20 bpm indicates tachypnea (abnormally rapid breathing). Respiratory rates between 21-24 bpm are considered mildly elevated, 25-29 bpm moderately elevated, and 30 bpm or above severely elevated, often signaling respiratory distress or failure (Badawy et al., 2017). Pediatric patients have different normal ranges: infants typically breathe 30-60 bpm, toddlers 24-40 bpm, and school-age children 18-30 bpm, with rates gradually decreasing to adult levels during adolescence.

In the context of triage, respiratory rate serves as a critical indicator of patient acuity and urgency. Abnormal respiratory rates often signal serious underlying conditions requiring immediate intervention. Tachypnea may indicate respiratory infections, pulmonary embolism, metabolic acidosis, sepsis, or cardiac dysfunction, while bradypnea can suggest central nervous system depression, drug overdose, or head injury (Philip et al., 2015). The Emergency Severity Index (ESI) algorithm specifically incorporates respiratory rate as a key factor in determining triage levels, with rates above 20 bpm potentially triggering high-priority categorization. Furthermore, respiratory rate is often the first vital sign to change in deteriorating patients, making it an early warning sign for clinical deterioration (Subbe et al., 2003).

Accurate respiratory rate measurement and documentation are essential for effective triage and patient monitoring. Despite its importance, respiratory rate is frequently under-assessed or estimated rather than measured directly, which can lead to missed indicators of patient deterioration (Cretikos et al., 2008). Healthcare providers should ensure that respiratory rate is systematically measured and recorded as part of routine vital sign assessment,

contributing to comprehensive triage evaluation and improved patient outcomes in emergency care settings.

References

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